Direct variation

1 If $(4, 512)$ is on the curve, $P = kQ^4$, solve for k.

2 A variable M is directly proportional to the square of N.

a Using k as the constant of proportionality, form an equation for M in terms of N.

b Given that $(4, 64)$ satisfies the equation, solve for k.

3 Consider the relationship where y is directly proportional to the cube of x.

a Using k as a constant of proportionality, state the equation relating x and y.

b The following table of values shows the relationship between x and y:

x	1	2	3	4	5
y	3	24	81	192	375

Solve for k, the constant of proportionality.

4 y is a power function which is directly proportional to x^n.

a Using k as the proportionality constant, form an equation for y in terms of x.

b The table below shows values that satisfy the relationship $y = kx^n$, for the case when $n = 2$.

x	2	4	6	8	10
y	32	128	288	512	800

Solve for the constant of proportionality k. Round your answer to the nearest integer.

5 y is a power function which is directly proportional to x^n.

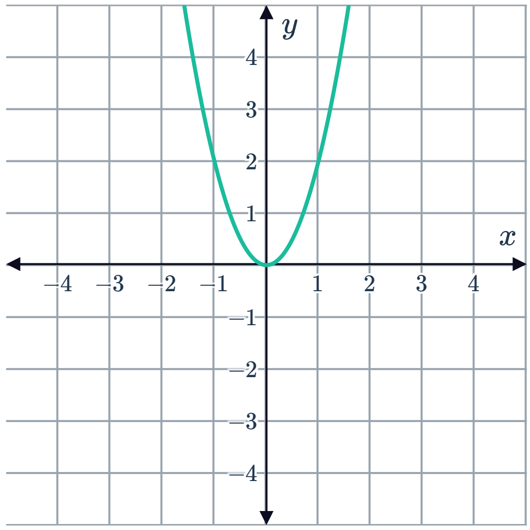
a Using k as the proportionality constant, form an equation for y in terms of x.

b The table below shows values that satisfy the relationship $y = kx^n$, for the case when $n = 3$.

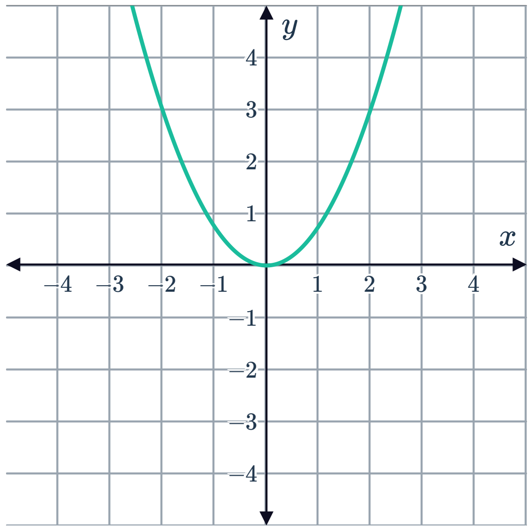
x	2	4	6	8	10
y	-48	-384	-1296	-3072	-6000

Solve for the constant of proportionality k. Round your answer to the nearest integer.

6 For the quadratic function pictured here, determine the constant of proportionality k:



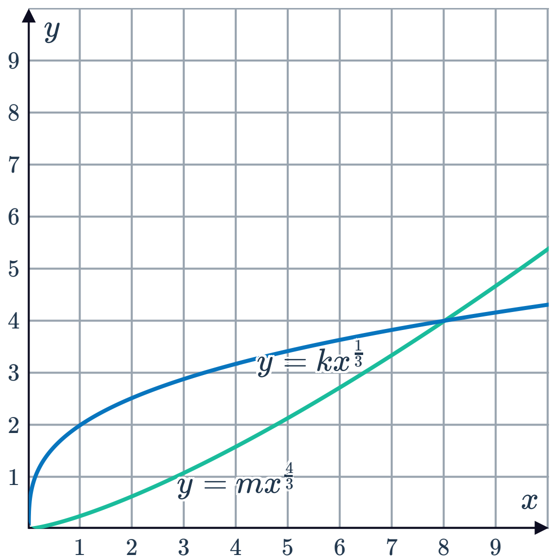
7 For the quadratic shown, determine the constant of proportionality k:



8 The following graph shows the intersection of the two functions $y = mx^{\frac{4}{3}}$ and $y = kx^{\frac{1}{3}}$:

a Find the value of m.

b Find the value of k.



Applications

9 In exchange rate calculations the constant of proportionality is the exchange rate itself. Determine the exchange rate, k, if someone can purchase:

a 889.00 AUD with 700.00 USD.

b 553.00 USD with 700.00 AUD.

10 The heart mass H of a mammal is proportional to its body mass M.

a Using k as the constant of proportionality, form an equation relating H and M.

b Given that a human with a body mass of 77kg has a heart mass of 0.462kg, determine k, the constant of proportionality.

c If a Chimpanzee has a body mass of 57kg, what is the heart mass H?

d If a Macaque has a heart mass of 0.084kg, solve for its body mass M.

11 The blood circulation time C of a mammal is proportional to the fourth root of its body mass B.

a Using k as the constant of proportionality, form an equation relating C and B.

b Given that an elephant with a body mass of 5290 kg has a blood circulation time of 148 seconds, determine k, the constant of proportionality. Round your answer to one decimal place.

c If a Orangutang has a body mass of 40 kg, find C, its circulation time. Round your answer to one decimal place.

d If a Squirrel has a circulation time of 22 seconds, what is the body mass? Round your answer to one decimal place.